

it suffices to show that, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $t \geq g - \delta$,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \left[\hat{\psi}_{g,t,\delta}^{dr,nev}(W_i; \hat{\kappa}_{g,t}^{nev}) - \psi_{g,t,\delta}^{dr,nev}(W_i; \kappa_{g,t}^{*,nev}) \right] = o_{p^*}(1), \quad (\text{A.8})$$

where $\kappa_{g,t}^{*,nev} = (\pi_g^{*'}, \beta_{g,t,\delta}^{*,nev})'$ is the vector of pseudo-true finite-dimensional parameters.

Towards this end, recall that

$$\hat{\psi}_{g,t,\delta}^{dr,nev}(W_i; \hat{\kappa}_{g,t}^{nev}) = \hat{\psi}_{g,t,\delta}^{treat,nev}(W_i; \hat{\beta}_{g,t,\delta}^{nev}) - \hat{\psi}_{g,t,\delta}^{comp,nev}(W_i; \hat{\pi}_g, \hat{\beta}_{g,t,\delta}^{nev}) - \hat{\psi}_{g,t,\delta}^{est,nev}(W_i; \hat{\pi}_g, \hat{\beta}_{g,t,\delta}^{nev})$$

where

$$\begin{aligned} \hat{\psi}_{g,t,\delta}^{treat,nev}(W; \hat{\beta}_{g,t,\delta}^{nev}) &= \hat{w}_g^{treat} \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\hat{\beta}_{g,t,\delta}^{nev}) \right) \\ &\quad - \hat{w}_g^{treat} \cdot \mathbb{E}_n \left[\hat{w}_g^{treat} \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\hat{\beta}_{g,t,\delta}^{nev}) \right) \right], \\ \hat{\psi}_{g,t,\delta}^{comp,nev}(W; \hat{\pi}_g, \hat{\beta}_{g,t,\delta}^{nev}) &= \hat{w}_g^{comp,nev}(\hat{\pi}_g) \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\hat{\beta}_{g,t,\delta}^{nev}) \right) \\ &\quad - \hat{w}_g^{comp}(\hat{\pi}_g) \cdot \mathbb{E}_n \left[\hat{w}_g^{comp}(\hat{\pi}_g) \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\hat{\beta}_{g,t,\delta}^{nev}) \right) \right], \\ \hat{\psi}_{g,t,\delta}^{est,nev}(W; \hat{\pi}_g, \hat{\beta}_{g,t,\delta}^{nev}) &= l_{g,t}^{or,nev}(\hat{\beta}_{g,t,\delta}^{nev})' \cdot \hat{M}_{g,t,\delta}^{dr,nev,1} + l_g^{ps,nev}(\hat{\pi}_g)' \cdot \hat{M}_{g,t,\delta}^{dr,nev,2}, \end{aligned}$$

with

$$\hat{w}_g^{treat} = \frac{G_g}{\mathbb{E}_n[G_g]}, \quad \hat{w}_g^{comp,nev}(\hat{\pi}_g) = \frac{\frac{\hat{p}_g(X; \hat{\pi}_g) C}{1 - \hat{p}_g(X; \hat{\pi}_g)}}{\mathbb{E}_n \left[\frac{\hat{p}_g(X; \hat{\pi}_g) C}{1 - \hat{p}_g(X; \hat{\pi}_g)} \right]},$$

and

$$\begin{aligned} \hat{M}_{g,t,\delta}^{dr,nev,1} &= \mathbb{E}_n \left[(\hat{w}_g^{treat} - \hat{w}_g^{comp,nev}(\hat{\pi}_g)) \cdot \dot{m}_{g,t,\delta}^{nev}(\hat{\beta}_{g,t,\delta}^{nev}) \right], \\ \hat{M}_{g,t,\delta}^{dr,nev,2} &= \mathbb{E}_n \left[\hat{\alpha}_g^{ps,nev}(\hat{\pi}_g) \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\hat{\beta}_{g,t,\delta}^{nev}) \right) \cdot \dot{p}_g(\hat{\pi}_g) \right] \\ &\quad - \mathbb{E}_n \left[\hat{\alpha}_g^{ps,nev}(\hat{\pi}_g) \cdot \hat{w}_g^{comp,nev}(\hat{\pi}_g) \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\hat{\beta}_{g,t,\delta}^{nev}) \right) \cdot \dot{p}_g(\hat{\pi}_g) \right], \\ \hat{\alpha}_g^{ps,nev}(\hat{\pi}_g) &= \frac{C}{(1 - p_g(X; \hat{\pi}_g))^2} \Bigg/ \mathbb{E}_n \left[\frac{p_g(X; \hat{\pi}_g) C}{1 - p_g(X; \hat{\pi}_g)} \right]. \end{aligned}$$

We first show that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \left(\hat{\psi}_{g,t,\delta}^{treat,nev}(W_i; \hat{\beta}_{g,t,\delta}^{nev}) - \psi_{g,t,\delta}^{treat,nev}(W_i; \beta_{g,t,\delta}^{*,nev}) \right) = o_{p^*}(1). \quad (\text{A.9})$$

Using the mean-value theorem, we write

$$\begin{aligned} &\frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \hat{\psi}_{g,t,\delta}^{treat,nev}(W_i; \hat{\beta}_{g,t,\delta}^{nev}) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \hat{w}_{g,i}^{treat} \cdot \left(Y_{i,t} - Y_{i,g-\delta-1} - m_{g,t,\delta}^{nev}(W_i; \beta_{g,t,\delta}^{*,nev}) \right) \\ &\quad - \sqrt{n} \left(\hat{\beta}_{g,t,\delta}^{nev} - \beta_{g,t,\delta}^{*,nev} \right)' \frac{1}{n} \sum_{i=1}^n V_i \cdot \hat{w}_{g,i}^{treat} \cdot \dot{m}_{g,t,\delta}^{nev}(W_i; \bar{\beta}_{g,t,\delta}^{nev}) \end{aligned}$$

$$\begin{aligned}
& - \mathbb{E}_n \left[\hat{w}_g^{treat} \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev} \left(\hat{\beta}_{g,t,\delta}^{nev} \right) \right) \right] \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \hat{w}_{g,i}^{treat} \\
& = \hat{I}_{g,t,\delta}^{1,treat} - \hat{I}_{g,t,\delta}^{2,treat} - \hat{I}_{g,t,\delta}^{3,treat},
\end{aligned}$$

where $\bar{\beta}_{g,t,\delta}^{nev}$ is an intermediate point that satisfies $\left| \bar{\beta}_{g,t,\delta}^{nev} - \beta_{g,t,\delta}^{*,nev} \right| \leq \left| \hat{\beta}_{g,t,\delta}^{nev} - \beta_{g,t,\delta}^{*,nev} \right|$ a.s.. From the strong law of large numbers and the fact that V is mean zero, independent of W , it follows that

$$\begin{aligned}
\hat{I}_{g,t,\delta}^{1,treat} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot w_{g,i}^{treat} \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev} \left(W_i; \beta_{g,t,\delta}^{*,nev} \right) \right) + o_p^*(1) \\
\hat{I}_{g,t,\delta}^{3,treat} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot w_{g,i}^{treat} \cdot \mathbb{E} \left[w_g^{treat} \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev} \left(\beta_{g,t,\delta}^{*,nev} \right) \right) \right] + o_p^*(1)
\end{aligned}$$

Similarly, from Assumptions 7 and 8, and the strong law of large numbers, we conclude that $\hat{I}_{g,t,\delta}^{2,treat} = o_p^*(1)$. Now (A.9) follows from combining these results.

Next, we show

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \left(\hat{\psi}_{g,t,\delta}^{comp,nev} (W_i; \hat{\pi}_g, \hat{\beta}_{g,t,\delta}^{nev}) - \psi_{g,t,\delta}^{comp,nev} (W_i; \pi_g^*, \beta_{g,t,\delta}^{*,nev}) \right) = o_p^*(1). \quad (\text{A.10})$$

Again, by the mean value theorem, we write

$$\begin{aligned}
& \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \hat{\psi}_{g,t,\delta}^{comp,nev} (W_i; \hat{\pi}_g, \hat{\beta}_{g,t,\delta}^{nev}) \\
&= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \hat{w}_g^{comp,nev} (W_i; \hat{\pi}_g) \cdot \left(Y_{i,t} - Y_{i,g-\delta-1} - m_{g,t,\delta}^{nev} \left(W_i; \beta_{g,t,\delta}^{*,nev} \right) \right) \\
& - \sqrt{n} \left(\hat{\beta}_{g,t,\delta}^{nev} - \beta_{g,t,\delta}^{*,nev} \right)' \frac{1}{n} \sum_{i=1}^n V_i \cdot \hat{w}_g^{comp,nev} (W_i; \hat{\pi}_g) \cdot \dot{m}_{g,t,\delta}^{nev} (W_i; \bar{\beta}_{g,t,\delta}^{nev}) \\
& - \mathbb{E}_n \left[\hat{w}_g^{comp,nev} (\hat{\pi}_g) \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev} \left(\hat{\beta}_{g,t,\delta}^{nev} \right) \right) \right] \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \hat{w}_g^{comp,nev} (W_i; \hat{\pi}_g) \\
&= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \frac{p_g (X_i; \pi_g^*) C_i}{\mathbb{E}_n \left[\frac{\hat{p}_g (X; \hat{\pi}_g) C}{1 - \hat{p}_g (X; \hat{\pi}_g)} \right]} \cdot \left(Y_{i,t} - Y_{i,g-\delta-1} - m_{g,t,\delta}^{nev} \left(W_i; \beta_{g,t,\delta}^{*,nev} \right) \right) \\
& + \sqrt{n} \left(\hat{\pi}_g - \pi_g^* \right)' \frac{1}{n} \sum_{i=1}^n V_i \cdot \frac{C_i}{\mathbb{E}_n \left[\frac{\hat{p}_g (X; \hat{\pi}_g) C}{1 - \hat{p}_g (X; \hat{\pi}_g)} \right]} \cdot \left(Y_{i,t} - Y_{i,g-\delta-1} - m_{g,t,\delta}^{nev} \left(W_i; \beta_{g,t,\delta}^{*,nev} \right) \right) \cdot \dot{p}_g (X_i; \bar{\pi}_g) \\
& - \sqrt{n} \left(\hat{\beta}_{g,t,\delta}^{nev} - \beta_{g,t,\delta}^{*,nev} \right)' \frac{1}{n} \sum_{i=1}^n V_i \cdot \hat{w}_g^{comp,nev} (W_i; \hat{\pi}_g) \cdot \dot{m}_{g,t,\delta}^{nev} (W_i; \bar{\beta}_{g,t,\delta}^{nev}) \\
& - \hat{M}_{g,t,\delta}^{comp} \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \frac{p_g (X_i; \pi_g^*) C_i}{\mathbb{E}_n \left[\frac{\hat{p}_g (X; \hat{\pi}_g) C}{1 - \hat{p}_g (X; \hat{\pi}_g)} \right]} \\
& - \hat{M}_{g,t,\delta}^{comp} \sqrt{n} \left(\hat{\pi}_g - \pi_g^* \right)' \frac{1}{n} \sum_{i=1}^n V_i \cdot \frac{C_i}{\mathbb{E}_n \left[\frac{\hat{p}_g (X; \hat{\pi}_g) C}{1 - \hat{p}_g (X; \hat{\pi}_g)} \right]} \cdot \dot{p}_g (X_i; \bar{\pi}_g)
\end{aligned}$$

$$= \widehat{I}_{g,t,\delta}^{1,comp} + \widehat{I}_{g,t,\delta}^{2,comp} - \widehat{I}_{g,t,\delta}^{3,comp} - \widehat{I}_{g,t,\delta}^{4,comp} - \widehat{I}_{g,t,\delta}^{5,comp},$$

where $\widehat{\beta}_{g,t,\delta}^{nev}$ and $\widehat{\pi}_g$ are intermediate points that satisfies $\left| \widehat{\beta}_{g,t,\delta}^{nev} - \beta_{g,t,\delta}^{*,nev} \right| \leq \left| \widehat{\beta}_{g,t,\delta}^{nev} - \beta_{g,t,\delta}^{*,nev} \right|$ a.s. and $\left| \widehat{\pi}_g - \pi_g^* \right| \leq \left| \widehat{\pi}_g - \pi_g^* \right|$ a.s., respectively, and

$$\widehat{M}_{g,t,\delta}^{comp} = \mathbb{E}_n \left[\widehat{w}_g^{comp,nev}(\widehat{\pi}_g) \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\widehat{\beta}_{g,t,\delta}^{nev}) \right) \right].$$

From the strong law of large numbers and the fact that V is mean zero, has variance one, and is independent of W , it follows that

$$\begin{aligned} \widehat{I}_{g,t,\delta}^{1,comp} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot w_g^{comp,nev}(W_i; \pi_g^*) \cdot \left(Y_{i,t} - Y_{i,g-\delta-1} - m_{g,t,\delta}^{nev}(W_i; \beta_{g,t,\delta}^{*,nev}) \right) + o_{p^*}(1), \\ \widehat{I}_{g,t,\delta}^{3,treat} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot w_g^{comp,nev}(W_i; \pi_g^*) \cdot \mathbb{E} \left[w_g^{comp} \cdot \left(Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{*,nev}) \right) \right] + o_{p^*}(1). \end{aligned}$$

Similarly, from Assumptions 7 and 8, and the strong law of large numbers, we conclude that

$$\widehat{I}_{g,t,\delta}^{2,comp} = \widehat{I}_{g,t,\delta}^{4,comp} = \widehat{I}_{g,t,\delta}^{5,comp} = o_{p^*}(1).$$

Now (A.10) follows from combining these results.

Next, we show that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \left(\widehat{\psi}_{g,t}^{est,nev}(W_i; \widehat{\pi}_g, \widehat{\beta}_{g,t,\delta}^{nev}) - \psi_{g,t}^{est,nev}(W_i; \pi_g^*, \beta_{g,t,\delta}^{*,nev}) \right) = o_{p^*}(1). \quad (\text{A.11})$$

From the strong law of large numbers and Assumptions 7 and 8,

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \widehat{\psi}_{g,t}^{est,nev}(W_i; \widehat{\pi}_g, \widehat{\beta}_{g,t,\delta}^{nev}) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \left(l_{g,t}^{or,nev}(W_i; \widehat{\beta}_{g,t,\delta}^{nev})' \cdot M_{g,t,\delta}^{dr,nev,1} + l_g^{ps,nev}(W_i; \widehat{\pi}_g)' \cdot M_{g,t,\delta}^{dr,nev,2} \right) + o_{p^*}(1) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot l_{g,t}^{est}(W_i; \widehat{\kappa}_{g,t}^{nev}) + o_{p^*}(1), \end{aligned}$$

where, for a generic $\kappa_{g,t}^{nev} = (\pi_g', \beta_{g,t,\delta}^{nev'})'$,

$$l_{g,t}^{est}(W; \kappa_{g,t}^{nev}) = l_{g,t}^{or,nev}(W; \beta_{g,t,\delta}^{nev})' \cdot M_{g,t,\delta}^{dr,nev,1} + l_g^{ps,nev}(W; \pi_g)' \cdot M_{g,t,\delta}^{dr,nev,2}$$

Thus, from Lemma 4.3 in Newey and McFadden (1994) and Assumption 7, it follows that

$$\begin{aligned} & Var^* \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n V_i \cdot \left(l_{g,t}^{est}(W_i; \widehat{\kappa}_{g,t}^{nev}) - l_{g,t}^{est}(W_i; \kappa_{g,t,\delta}^{*,nev}) \right) \right) \\ &= \frac{1}{n} \sum_{i=1}^n \left(l_{g,t}^{est}(W_i; \widehat{\kappa}_{g,t}^{nev}) - l_{g,t}^{est}(W_i; \kappa_{g,t,\delta}^{*,nev}) \right)^2 \\ &= o_p(1), \end{aligned}$$

which, in turn, implies (A.11).

Taking (A.9), (A.10), and (A.11) together, we then establish (A.8). Thus, by (A.7), we have

$$\sqrt{n} \left(\widehat{ATT}_{t \geq (g-\delta)}^{*,dr,nev} - \widehat{ATT}_{t \geq (g-\delta)}^{dr,nev} \right) \xrightarrow[*]{d} N(0, \Sigma).$$

Finally, by the continuous mapping theorem, see e.g. Theorem 10.8 in Kosorok (2008), for any continuous functional $\Gamma(\cdot)$

$$\Gamma \left(\sqrt{n} \left(\widehat{ATT}_{t \geq (g-\delta)}^{*,dr,nev} - \widehat{ATT}_{t \geq (g-\delta)}^{dr,nev} \right) \right) \xrightarrow[*]{d} \Gamma(N(0, \Sigma)),$$

concluding our proof. $\square$

Appendix B: Additional Results for Repeated Cross Sections

In this section we extend our identification results to the case where one has access to repeated cross sections data instead of panel data. Here we assume that for each unit in the pooled sample, we observe $(Y, G_2, \dots, G_{\mathcal{T}}, C, T, X)$ where $T \in \{1, \dots, \mathcal{T}\}$ denotes the time period when that unit is observed. Let $T_t = 1$ if an observation is observed at time t , and zero otherwise.

We assume that random samples are available for each time period.

Assumption B.1. *Conditional of $T = t$, the data are independent and identically distributed from the distribution of $(Y_t, G_2, \dots, G_{\mathcal{T}}, C, X)$, for all $t = 1, \dots, \mathcal{T}$, with $(G_2, \dots, G_{\mathcal{T}}, C, X)$ being invariant to T .*

Assumption B.1 implies that our sample consists of random draws from the mixture distribution

$$F_M(y, g_2, \dots, g_{\mathcal{T}}, c, t, x) = \sum_{t=1}^{\mathcal{T}} \lambda_t \cdot F_{Y, G_2, \dots, G_{\mathcal{T}}, C, X|T}(y, g_2, \dots, g_{\mathcal{T}}, c, x|t),$$

where $\lambda_t = P(T_t = 1)$. It also rules-out compositional changes across time. This assumption is related to the sampling assumption imposed by Abadie (2005) and Sant'Anna and Zhao (2020) in the two periods, two groups DiD setup. Notice that, once one conditions on the time period, then expectations under the mixture distribution correspond to population expectations. Also, because X , G_g , and C are observed for all units, by the stationarity condition one can use draws from the mixture distribution to estimate the generalized propensity score. With some abuse of notation, we then use $p_{g,s}(X)$ as a short notation for $\mathbb{E}_M[G_g|X, G_g + (1 - D_s)(1 - G_g) = 1]$, where $\mathbb{E}_M[\cdot]$ denotes expectations with respect to $F_M(\cdot)$. Also, we use $p_g(X) = p_{g,\mathcal{T}}(X) = \mathbb{E}_M[G_g|X, G_g + C = 1]$.

Before formalizing all the results, we need to introduce some additional notation. Let $m_{c,t}^{rc,nev}(x) \equiv \mathbb{E}_M[Y|X = x, C = 1, T = t]$, $m_{g,t}^{rc,treat}(x) \equiv \mathbb{E}_M[Y|X = x, G_g = 1, T = t]$ and $m_{s,t}^{rc,ny}(x) \equiv \mathbb{E}_M[Y|X = x, D_s = 0, G_g = 0, T = t]$. Consider the weights

$$\begin{aligned} w^{treat}(a, b) &= T_b \cdot G_a / \mathbb{E}_M[T_b \cdot G_a], \\ w_{nev}^{comp}(a, b) &= \frac{T_b \cdot p_a(X) C}{1 - p_a(X)} \bigg/ \mathbb{E}_M \left[\frac{T_b \cdot p_a(X) C}{1 - p_a(X)} \right], \\ w_{ny}^{comp}(a, b, s) &= \frac{T_b \cdot p_{a,s}(X) (1 - D_b) (1 - G_a)}{1 - p_{a,s}(X)} \bigg/ \mathbb{E}_M \left[\frac{T_b \cdot p_{a,s}(X) (1 - D_b) (1 - G_a)}{1 - p_{a,s}(X)} \right]. \end{aligned}$$

Finally, consider the outcome regression (OR) estimands,

$$ATT_{or,rc}^{nev}(g, t; \delta) = \mathbb{E}_M \left[\frac{G_g}{\mathbb{E}_M[G_g]} \left(\left(m_{g,t}^{rc,treat}(X) - m_{g,g-\delta-1}^{rc,treat}(X) \right) - \left(m_{c,t}^{rc,nev}(X) - m_{c,g-\delta-1}^{rc,nev}(X) \right) \right) \right],$$